Paper2Video: Automatic Video Generation from Scientific Papers

Zeyu Zhu*, Kevin Qinghong Lin*, Mike Zheng Shou†
Show Lab, National University of Singapore

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Motivation: Need for Automated Academic Presentation Videos

- Academic presentation videos critical for research dissemination.
- Manual production is highly **labor-intensive**, requiring hours for just a few minutes.
- Unique challenges:
  - Long-context multi-modal input from dense scientific papers.
  - Multi-channel alignment: slides, subtitles, speech, talking-head, cursor.
  - No existing benchmarks or tailored metrics to evaluate

Related Work: Landscape and Challenges

- **Natural Video Generation:**
  - Diffusion models generate short videos from text [**?**, **?**].
  - Long and multi-shot videos remain challenging [**?**].
- **Multi-agent frameworks for videos:**
  - MovieAgent generates movies via role decomposition [**?**].
  - PresentAgent combines slide and speech but lacks personalization and academic slide quality [**?**].
- **AI for Research:**
  - Automatic generation of scientific posters [**?**] and slides [**?**].
  - Few efforts on **academic presentation video synthesis**.
- **Our work fills this gap**, focusing on long, multimodal, personalized academic video generation with comprehensive evaluation.

Method Overview: The PaperTalker Agent

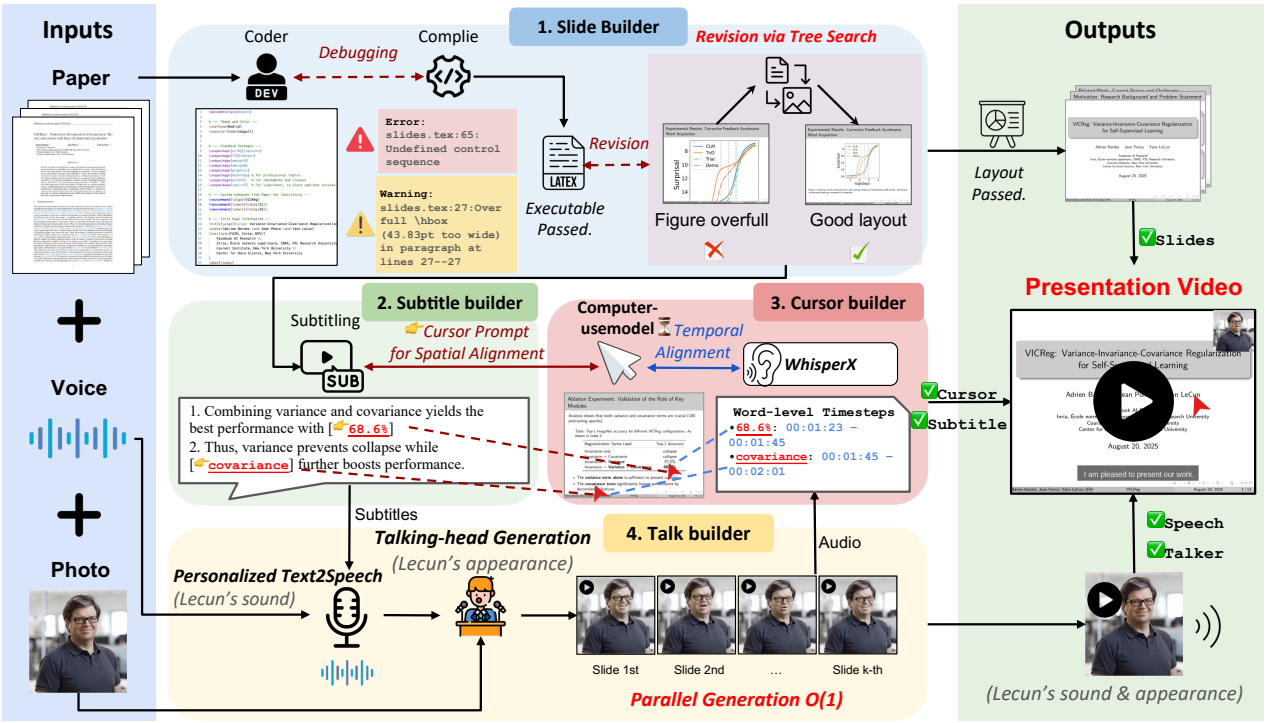


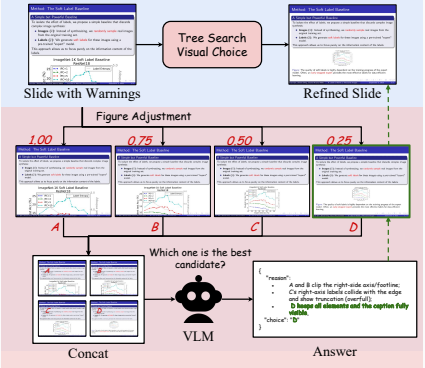
Figure 1: Pipeline of **PaperTalker**: Modular multi-agent approach integrating slide generation, subtitle/cursor creation, and talking-head synthesis.

Key Modules

- **Slide Builder**: Generates LaTeX Beamer slides, refines layouts via compilation feedback.
- **Subtitle & Cursor Builder**: Produces aligned subtitles with spatially grounded cursor trajectories.

Method: Slide Builder and Layout Refinement

- **Slides are generated directly** from the paper's LaTeX source using Beamer for professional academic layouts.
- Compilation feedback (errors, warnings) used to iteratively fix code ensuring correctness.
- **Tree Search Visual Choice:**
  - Rule-based generation of layout variants by adjusting font and figure scales.
  - Use vision-language model to **select the best candidate visually**.
  - Solves overflow and layout issues efficiently beyond parameter tuning by LLMs.



Method: Subtitle and Cursor Builder

- Subtitles are generated for each slide with a vision-language model.
- **Visual focus prompts** highlight key content for each subtitle sentence.
- Cursor trajectory is **spatially aligned** per sentence using GUI-grounding models.
- **Temporal alignment** with narration timing achieved using WhisperX.

Benefit

- **Enhanced audience guidance** by highlighting the precise focus of the narration on the slides.
- Improves accessibility and content grounding.

Method: Talker Builder and Parallelization

- Personalize speech synthesis using author voice sample with F5-TTS at slide granularity.
- Talking-head video generated based on author portrait using FantasyTalking and Hallo2.
- **Slide-wise parallel generation** for significant acceleration (6× speedup).
- You save hours of generation time while preserving video quality and engagement.

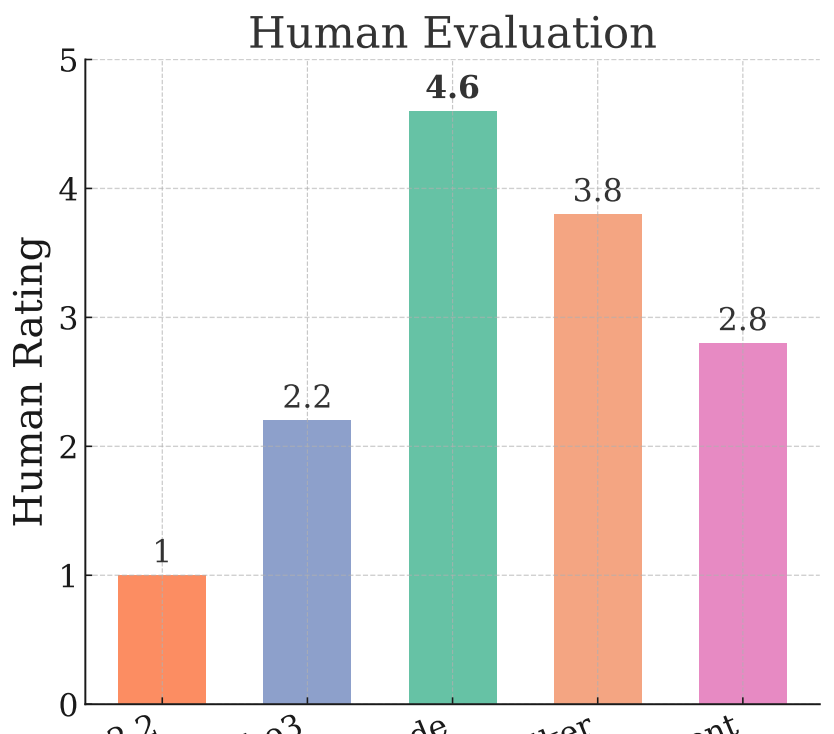
Experimental Settings

- **Baselines:**
  - End-to-end video generation models (e.g., Wan2.2, Veo3).
  - Multi-agent frameworks (e.g., PresentAgent).
  - Human-created presentations as upper bound.
- **Dataset:** Paper2Video benchmark with 101 paper-video pairs from recent AI conferences.
- Hardware: 8× NVIDIA RTX A6000 GPUs.
- Evaluation Models:
  - Vision-language model: GPT-4.1.
  - Video language model: Gemini-2.5-Flash.

Experimental Results: Main Evaluation

Table 1: Main evaluation results on Paper2Video benchmark. Note: PaperTalker* is without presenter/cursor.

Method	Sim. (Speech/Content)	Arena Win	PresentQuiz	IP Mem.	Duration (s)
HumanMade	**1.00 / 5.00**	**50%**	0.738 / 0.908	-	375
Wan2.2	NA / NA	1.1%	0.251 / 0.551	11.5%	4
Veo3	0.133 / NA	1.2%	0.367 / 0.585	31.3%	8
PresentAgent	0.045 / 1.47	2.0%	0.548 / 0.654	12.5%	430
PaperTalker*	**0.646 / 1.97**	15.2%	0.835 / 0.949	37.5%	234
[illegible]	[illegible]	**17.0%**	**0.842 / 0.951**	**50.0%**	234



- **PaperTalker** outperforms all baselines in both automatic and human evaluations.
- Presenter video boosts engagement and quality preference by approx. 10%.

Experimental Results: Efficiency and Cost

Table 2: Generation cost for each method.

Method	Token (K)↓	Time (min)↓	Cost ($)↓
Veo3 [?]	NA	**0.4**	1.667
Wan2.2 [?]	NA	8.1	0.280
PresentAgent [?]	241	39.5	0.003
PaperTalker (w/o Talker)	**62**	15.6	**0.001**
PaperTalker (w/o Parallel)	**62**	287.2	**0.001**
PaperTalker	**62**	**48.1 (6× faster)**	**0.001**

- Parallel generation yields **6× speedup** vs. serial.
- PaperTalker achieves **lowest generation cost** among compared methods.

Ablation Study: Impact of Cursor and Tree Search Modules

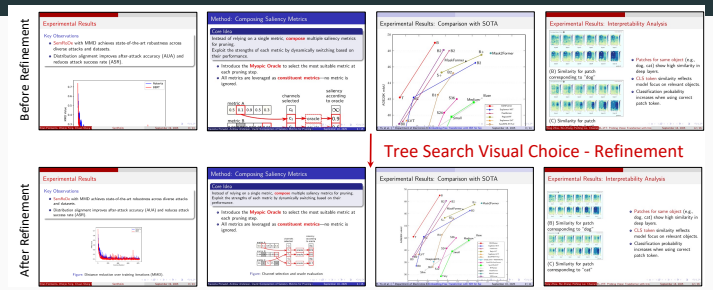


Figure 4: Slides before (top row) and after (bottom row) Tree Search Visual Choice layout refinement.

- **Cursor Highlight**: Improves content localization, increasing QA accuracy from 8% to 63%.
- **Tree Search Visual Choice**: Fixes layout overflows and improves slide design and coherence.

Table: Ablation on Cursor

Method	Localization Accuracy
Without Cursor	0.084
With Cursor	**0.633**

Limitations and Future Research Directions

- **Current Limitations:**
  - Difficulty in parsing long scientific documents with complex equations end-to-end.
  - Some layouts and slide designs still need refinement.
  - Talking-head synthesis quality and naturalness can be improved further.
- **Future Directions:**
  - Real-time interactive presentation video agents enabling Q&A and dynamic tutorials.
  - Enhanced semantic understanding of longer, complex papers.
  - Support for multilingual presentations and accessibility features.
  - Closed-loop evaluation with human and AI-in-the-loop feedback.
  - Full open-sourcing for community collaboration.

Thank You!

Questions and Discussion Welcome

e1374425@nus.edu.sg